

Education

- 2022–Present **PhD, Computer Science, Vanderbilt University.**
Advisor: Prof. Soheil Kolouri. Expected 2026.
- 2024 **MS, Computer Science, Vanderbilt University** (awarded en route to PhD).
- 2016–2020 **BS, Electrical Engineering, University of Tehran.**

Research Summary

Post-training methods that make large models cheaper to run: low-rank factorization (ZS-SVD, IO-SVD), activation quantization (ConQuR), and expert selection for MoE serving (BASE). Earlier work covers compact reparameterizations (PRANC, MCNC) and dataset distillation.

Publications [\[Google Scholar Link\]](#) (* denotes equal contribution)

Main Conference Proceedings

ConQuR: Corner Aligned Activation Quantization via Optimized Rotations for LLMs. C. Thrash, **A. Abbasi**, S. Kolouri. NeurIPS 2026. [\[link\]](#)

LLM activation quantization learning optimized rotations that align activations with hypercube corners; 19.2× faster quantization than SpinQuant with lower perplexity across Llama families.

Zero Sum SVD: Balancing Loss Sensitivity for Low Rank LLM Compression. **A. Abbasi**, C. Thrash*, H. Qin*, S. Sharma, S. Seifi, S. Kolouri. ICML 2026. [\[link\]](#)

Selects singular components across all layers with a zero-sum rule on first-order loss estimates; at 40% retention, 5.86× the uncompressed throughput and 16% lower WikiText-2 perplexity than SVD-LLM.

MCNC: Manifold-Constrained Reparameterization for Neural Compression. C. Thrash, **A. Abbasi**, R. Andreas, P. Nooralinejad, S. A. Koohpayegani, H. Pirsiavash, S. Kolouri. ICLR 2025. [\[link\]](#)

Constrains parameters to a low-dimensional nonlinear manifold generated by a frozen random network; +22.3 accuracy points over PLATON pruning at 50× compression (ViT-Ti, ImageNet-100).

BrainWash: A Poisoning Attack to Forget in Continual Learning. **A. Abbasi**, P. Nooralinejad, H. Pirsiavash, S. Kolouri. CVPR 2024. [\[link\]](#)

Untargeted poisoning attack forcing forgetting in continual learners via model inversion of past data; up to 80% more forgetting than uniform noise.

PRANC: Pseudo Random Networks for Compacting Deep Models. P. Nooralinejad, **A. Abbasi**, S. A. Koohpayegani*, K. Pourahmadi Meibodi*, R. M. S. Khan*, S. Kolouri, H. Pirsiavash. ICCV 2023. [\[link\]](#)

Reparameterizes weights as a linear mixture of frozen random networks regenerated from a seed; +8.1 accuracy points over HashedNet at about 100× compression (ResNet-18, ImageNet-100).

Preprints

BASE: Batch-Aware Selection of Experts Using Predicted Removal Error for Efficient MoE Decoding. **A. Abbasi**, J. Shi, S. Kolouri. arXiv 2026. [\[link\]](#)

Batch-aware MoE expert selection from predicted removal error, with custom GPU kernels; +29.5 accuracy points over the strongest baseline, or 60% faster than dense within 0.4 points (Qwen3-30B-A3B).

IO-SVD: Input-Output Whitenened SVD for Adaptive-Rank LLM Compression. A. Abbasi, C. Thrash*, H. Qin*, H. Pirsiavash, S. Kolouri. arXiv 2026. [\[link\]](#)

LLM/VLM compression via KL-aware input-output whitening and adaptive ranks; +10.2 avg. accuracy points over WSVD (SmolVLM-2B); $4.34\times$ decode speedup with KV-cache compression (LLaMA-2-7B).

Vector-Quantized Soft Label Compression for Dataset Distillation. A. Abbasi, A. Shahbazi, H. Pirsiavash, S. Kolouri. arXiv 2026. [\[link\]](#)

Compresses soft labels stored with distilled datasets via a vector-quantized autoencoder; $30\text{--}40\times$ additional compression on ImageNet-1K while retaining over 90% of baseline accuracy.

Workshop Papers

One Category One Prompt: Dataset Distillation using Diffusion Models. A. Abbasi, A. Shahbazi, H. Pirsiavash, S. Kolouri. CVPR Workshops (SynData4CV) 2026. [\[link\]](#)

Distills each class into a learnable text prompt for a diffusion model; +12.6 accuracy points over RDED on ImageNet-1K at 50 images/class (ResNet-18, stored soft labels).

Diffusion-Augmented Coreset Expansion for Scalable Dataset Distillation. A. Abbasi, S. Imani, C. An, G. Mahalingam, H. Shrivastava, M. Diesendruck, H. Pirsiavash, P. Sharma, S. Kolouri. CVPR Workshops (SynData4CV) 2026. [\[link\]](#)

Expands compact low-resolution coresets with uncertainty-guided diffusion augmentation; +10.1 accuracy points over RDED on ImageNet-1K at 10 images/class (ResNet-18).

Industry Experience

Jun–Sep 2024 **Research Intern at Microsoft, Redmond, WA.**

Developed an uncertainty-guided diffusion augmentation pipeline that expands compact low-resolution coresets; led to a CVPR 2026 workshop paper [\[link\]](#).

Honors & Service

2026 **ICML 2026** Gold Reviewer (top 25%)

Reviewer NeurIPS '23, '25, '26; ICLR '24, '26; ICML '26; CVPR '26; AAAI '26; WACV '26; IEEE TPAMI

2025 Compute grants: **Lambda, Google Cloud**

Teaching Experience

Spring 2023 TA, Advanced AI, **Vanderbilt University**

2018–2019 TA, Electrical Circuits I, Engineering Math, Probability & Statistics, **University of Tehran**

Skills

Python, C/C++, PyTorch, Transformers, vLLM, JAX, Diffusers, CUDA, CUTLASS, Nsight

References

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